

MICROCONTROLLER-BASED DC MOTOR CONTROLLER



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Motion control plays a vital role in industrial automation. Manufacturing plants in industries like chemical, pharmaceutical, plastic and textile, all require motion control. And it may be a flat-belt application, flow-control application or mixing of substances. Different types of motors—AC, DC, servo or stepper—are used depending upon the application. Of these, DC motors are widely used because controlling a DC motor is somewhat easier than other kinds of motors.

The motion of a DC motor is controlled using a DC drive. DC drive changes the speed and direction of motion of the motor. Some of the DC drives are just a rectifier with a series resistor that converts standard AC supply into DC and gives it to the motor through a switch and a series resistor to change the speed and direction of rotation of the motor. But many of the DC drives have an inbuilt microcontroller that provides programmable facilities, message display on LCD, precise control and also protection for motors. Using the DC drive you can program the motion of the motor, i.e., how it should rotate.

Here are some of the features of this DC motor controller:

1. Controlled through

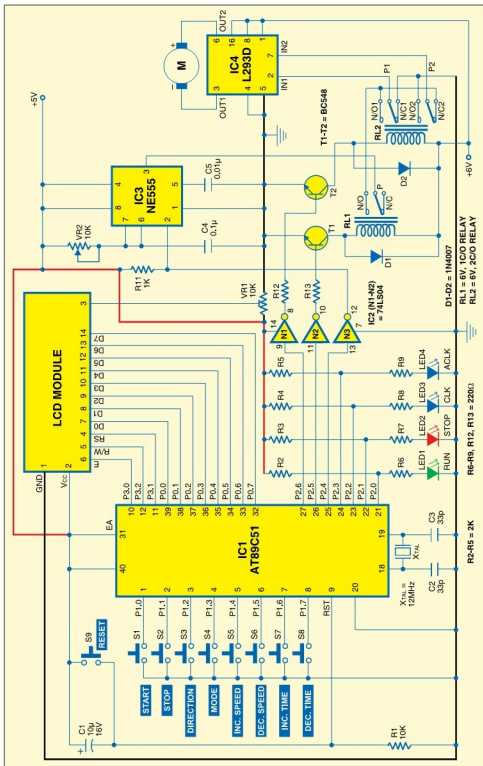


Fig. 1: Circuit of the microcontroller-based DC motor controller